

LAKAN SINGH DEGAVATH

AI/ML & SOFTWARE ENGINEER | GENERATIVE AI | LLM SYSTEMS | RAG | AGENTIC AI | DISTRIBUTED SYSTEMS

San Francisco, CA | [Gmail](#) | [Portfolio](#) | [GitHub](#) | [LinkedIn](#)

PROFESSIONAL SUMMARY

AI/ML and Software Engineer with 7 years of experience spanning production software, machine learning, Generative AI, LLM/RAG systems, agentic workflows, model evaluation, inference optimization, distributed backend systems, streaming data, and MLOps. Hands-on with Python, PyTorch, FastAPI, LangGraph/LangChain, AWS, Docker/Kubernetes, Kafka/Spark, C++ and .NET. Builds production-oriented systems backed by automated testing, measurable benchmarks, observability, and reliability engineering.

TECHNICAL SKILLS

AI / LLM Systems:	Generative AI, LLMs, RAG, Agentic AI, LangGraph, LangChain, Hugging Face Transformers, Embeddings, Semantic Search, Vector Databases, Prompt Engineering, Context Engineering.
ML / Deep Learning:	PyTorch, TensorFlow, JAX, Keras, scikit-learn, XGBoost, NLP, Machine Learning, Deep Learning, Model Evaluation, LLM Evaluation, Benchmarking, Failure-Mode Analysis, Edge-Case Testing.
Inference / Performance:	vLLM, SGLang, Model Serving, Quantization, Dynamic Batching, Low-Latency Inference, CPU/GPU Profiling, CUDA, GPU Acceleration, x86-64, ARM, Multicore Processing, Memory Optimization.
Backend / Systems:	Python, C++, C, Java, Scala, Go, C#, SQL, JavaScript/TypeScript, FastAPI, REST APIs, gRPC, Microservices, Asynchronous Programming, Concurrency, Distributed Systems, Fault Tolerance.
Data / Streaming:	PostgreSQL, Redis, MongoDB, FAISS, Pinecone, Weaviate, Pandas, NumPy, Spark, PySpark, Kafka, Spark Streaming, ETL, JSON/CSV, Data Validation.
Cloud / MLOps:	AWS, SageMaker, S3, EC2, Lambda, Docker, Kubernetes, MLflow, CI/CD, Jenkins, Experiment Tracking, Model Monitoring, Logging, Tracing, Observability, LLMops.
Software Engineering:	System Design, Scalable Backend Services, Git, GitHub, Linux, Testing, Unit Testing, Integration Testing, Debugging, Code Reviews, Agile/Scrum, Jira.

PROFESSIONAL EXPERIENCE

AI/ML Engineer | Client: EasyBee AI

United States | Apr 2026 - Present

- Build production-oriented LLM, RAG and agentic AI services using Python, FastAPI, LangGraph/LangChain, embeddings and vector retrieval, including planning, tool use, retrieval, memory and self-correction workflows.
- Develop low-latency inference and backend services with distributed-systems, concurrency and fault-tolerance patterns; profile CPU/GPU execution and apply batching, quantization and memory tuning to improve latency and throughput.
- Implement model/LLM evaluation, automated tests, edge-case and failure-mode analysis, monitoring and reliability workflows to improve production readiness.
- Build AWS-based ML infrastructure using SageMaker, S3, Lambda, Docker, CI/CD, experiment tracking and monitoring; develop PySpark and structured-data pipelines for AI workloads.

Software Developer | Client: EasyBee AI

United States | Feb 2026 - Apr 2026

- Developed enterprise application and reporting workflows using C#, .NET, Blazor, REST APIs, SQL-backed systems, data integration, and CI/CD-oriented engineering practices.
- Worked with SQL Server/Oracle-oriented data workflows, reporting APIs, data modeling and integration patterns aligned with enterprise application requirements.

Graduate Research Assistant - AI & Machine Learning | Avila University

United States | Apr 2024 - Dec 2025

- Built a reproducible Python NLP/ML pipeline for 205K+ product reviews covering preprocessing, feature engineering, training, evaluation and comparative analysis.
- Benchmarked Random Forest, Naive Bayes and SVM approaches against GPT-4 zero-shot classification using accuracy, precision, recall and F1; analyzed mixed-sentiment behavior and prediction errors.
- Extended experimentation to deep-learning/Transformer approaches and contributed to research presentation and co-authored academic work.

Software Engineer | Enterprise / Distributed Systems | Client: Shopify

Hyderabad, India | Sep 2020 - Dec 2023

- Engineered a real-time fraud-detection pipeline processing 3M+ financial transactions using machine-learning models, Kafka and Spark Streaming; improved inference latency by 45%.
- Evaluated XGBoost, Random Forest and Isolation Forest using ROC-AUC/PR metrics and explainability techniques including SHAP/LIME; reduced false positives to fewer than 2 per 1,000 transactions.
- Built backend and streaming components in Python/Java/Scala with production testing, observability and reliability practices; work contributed to preventing \$90K+ in annual fraud losses.

Associate Software Engineer | Enterprise Software Engineering | Client: Shopify

Hyderabad, India | Sep 2019 - Aug 2020

- Developed and tested C++/Java/Scala components using object-oriented design, code reviews and CMake-based build workflows.
- Analyzed C/C++ performance on x86-64 and multicore systems using profiling, cache-aware programming, multithreading and memory-management techniques.

PROJECTS

Enterprise Multi-Agent AI Platform | Python, FastAPI, LangGraph, FAISS, Docker, Pytest

- Engineered a production-oriented multi-agent AI platform with supervisor routing, RAG, tool execution, memory, bounded self-correction,

runtime evaluation, and observability; validated with 76 automated tests.

- Achieved Hit@4 1.00, Recall@4 1.00, and MRR 1.00 on a controlled synthetic enterprise-policy retrieval benchmark.

GPU-Accelerated LLM Inference Benchmark | PyTorch, Hugging Face, Apple MPS

- Built a reproducible LLM inference benchmarking framework for latency, throughput, batching, and device-aware execution; validated with 44 automated tests.
- Measured up to 604.88 tokens/sec and an 11.66× batch-1-to-batch-8 throughput increase using DistilGPT2/FP32 on Apple MPS.

Real-Time Fraud Detection Platform | Kafka, Spark, scikit-learn, PostgreSQL, FastAPI

- Developed a production-style streaming ML platform integrating Kafka, Spark, model inference, REST APIs, and PostgreSQL, with deterministic held-out synthetic evaluation and 12 automated tests.
- Achieved ROC-AUC 0.9947 and PR-AUC 0.8292 on a controlled small-scale synthetic benchmark.

Enterprise Reporting & Data Integration Platform | C#, .NET 10, ASP.NET Core, EF Core, Blazor

- Built an enterprise reporting platform with CSV ingestion, data validation, SQL-backed persistence, REST APIs, and reporting workflows using ASP.NET Core, EF Core, and Blazor.
- Containerized and CI-enabled the application with automated testing across core data-ingestion, validation, and reporting workflows.

EDUCATION

Master of Science in Computer Science - AI/ML Specialization | Avila University | GPA: 3.91/4.00